

ETX1100/5900 Business Statistics

Week 2

Probability: Concepts & Applications

Introduction to the Normal Distribution

Figure 1: ETX5900 source PDF, Week 2, page 1

Learning Outcomes for Week 2

Fundamentals of Probability

- Understand and calculate the concept of probability
- Interpret probabilities:
  - ✓ Marginal probability
  - ✓ Joint probability
  - ✓ Conditional probability
- Evaluate the relationship between variables

Introduction to the Normal Distribution

- Understand the concept of a continuous random variable
- Analyse the characteristics of a normal distribution
- Calculate and interpret probabilities for continuous random variables using Excel

Figure 2: ETX5900 source PDF, Week 2, page 2

Learning Outcomes for Week 2

- Understand the concept of probability
- Calculate and interpret probabilities:
  - Marginal probability
  - Joint probability
  - Conditional probability
- Apply the concepts of probability for categorical data analysis
- Evaluate the relationship between variables

Textbook references:
Berenson et al *Basic Business Statistics 5th edition,*
Chapter 4 Sections 4.1 and 4.2
Chapter 6 Sections 6.1-6.2

Figure 3: ETX5900 source PDF, Week 2, page 3

What is PROBABILITY?

- **Probability = quantification of chance / likelihood of an occurrence**

Probability statements:

✓ Event A has zero probability **P(A) = 0** → There is no chance that event A will occur

✓ Event B has probability of one **P(B)=1** → Event B will occur with 100% certainty

Probability of all possible events add to 1

- **Eg Experiment: Toss a fair coin**
- $\mathrm{P}(Head) = 0.5 \qquad \mathrm{P}(Tail) = 0.5$
- **What is the total probability?** $\boldsymbol{P(Head) + P(Tail) = 1}$

Figure 4: ETX5900 source PDF, Week 2, page 4

Intersection (Joint probability) & Union

Intersection (joint probability)

Helps us understand the nature of the relationship between two events

- P(A and B) or P$(A \cap B)$ - known as **"JOINT"** probability

Union

- P(A or B) or P$(A \cup B)$ describes either A OR B
- P$(A \text{ or } B) = P(A) + P(B) - \Pr(A \text{ and } B)$
- **Note: P(A) and P(B) are referred to as marginal probabilities**

Figure 5: ETX5900 source PDF, Week 2, page 5

Conditional Probability

Conditional Probability

Probability of an **event (A) given that (*conditional on*) another event (B)** having occurred.

P(A given B) or P (A|B) is known as *conditional probability*

$$\text{Conditional Probability} = \frac{\text{Joint probability of the two events}}{\text{Marginal probability of the condition}}$$

$$\mathbf{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

Figure 6: ETX5900 source PDF, Week 2, page 6

Mutually Exclusive and Independent Events

Mutually exclusive:

- **Definition:** Events that cannot occur together

We know that P(A or B) = P(A) + P(B) – P(A and B). For mutually exclusive events, P(A and B) = 0

➢ Example: Toss a fair coin P(Not Head) = P(Tail) = 1-P(Head)

Independent:

- **Definition:** Probability of one event occurring unaffected by the outcome of the other

Event A is independent of event B if:

P(A|B) = P(A) OR $P(A \cap B) = P(A) \times P(B)$

Figure 7: ETX5900 source PDF, Week 2, page 7

Frequency Table

Open"***Heart Disease"*** sheet in **Week 2 Workshop Data.xlsx.** It contains data on the diagnosis of heart disease and exercise levels. Generate 2 relative frequency tables

- Exercise level
- Medical condition

What information do the tables convey?

Figure 8: ETX5900 source PDF, Week 2, page 8

Instructions:

Exercise Level

1. Place the cursor in any cell within the data
2. Drag & drop **Exercise** under 1. Rows & 2. Values.
3. Right-click on **Exercise** entries > Value Field Setting >
4. Summarize value by > Count;
5. Show Values As > % of Column Total

Medical Condition

1. Place the cursor in any cell within the data
2. Drag & drop **Heart Disease** under 1. Rows & 2. Values.
3. Right-click on **Heart Disease** entries > Value Field Setting >
4. Summarize value by > Count;
5. Show Values As > % of Column Total

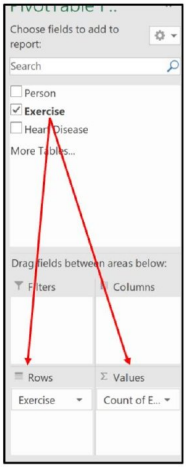


Figure 9: ETX5900 source PDF, Week 2, page 9

Relative Frequency Tables:

Exercise	Count of Exercise
Minimal	64.32%
Moderate	35.68%
Grand Total	**100.00%**

Heart Disease	Count of Heart Disease
No	94.74%
Yes	5.26%
Grand Total	**100.00%**

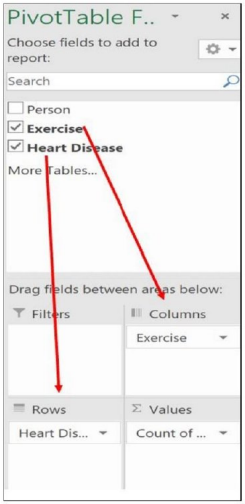
Figure 10: ETX5900 source PDF, Week 2, page 10

Calculating Joint Probabilities

Construct a two-way frequency table (contingency table/Pivot table) by placing

***Heart Disease* in rows and *Exercise* in Columns. Display as "count"**

Count of Heart Disease	Exercise Level		
Heart Disease	**Minimal**	**Moderate**	**Grand Total**
No	2996	1741	4737
Yes	220	43	263
Grand Total	3216	1784	5000



Convert to probabilities

Probabilities	Exercise Level		
Heart Disease	**Minimal**	**Moderate**	**Grand Total**
No			
Yes			
Grand Total			

Figure 11: ETX5900 source PDF, Week 2, page 11

Calculating Joint Probabilities: Manual Calculations

Construct a two-way frequency table (contingency table/Pivot table) by placing *Heart Disease* in rows and *Exercise* in Columns. Display as "count"

Count of Heart Disease	Exercise Level		
Heart Disease	**Minimal**	**Moderate**	**Grand Total**
No	2996	1741	4737
Yes	220	43	263
Grand Total	3216	1784	5000

Probabilities	Exercise Level		
Heart Disease	**Minimal**	**Moderate**	**Grand Total**
No	=2996/5000	=1741/5000	=4737/5000
Yes	=220/5000	=43/5000	=263/5000
Grand Total	**=3216/5000**	**=1784/5000**	**1**

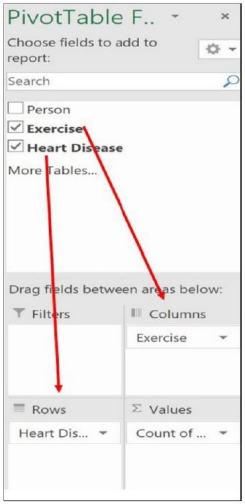


Figure 12: ETX5900 source PDF, Week 2, page 12

Pivot table: % of Grand Total

Construct a pivot table by placing *Heart Disease* in rows and *Exercise* in Columns. Display as "% of Grand Total"

Count of Heart Disease	Exercise Level		
Heart Disease	Minimal	Moderate	Grand Total
No	59.92%	34.82%	94.74%
Yes	4.40%	0.86%	5.26%
Grand Total	64.32%	35.68%	100.00%

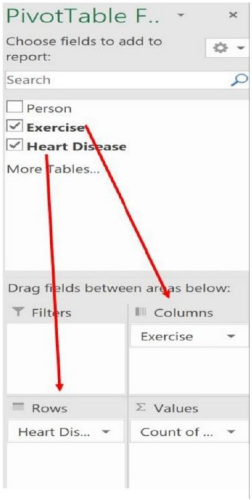


Figure 13: ETX5900 source PDF, Week 2, page 13

Task 1: Heart Disease and Exercise Level

Count of Heart Disease	Exercise Level		
Heart Disease	Minimal	Moderate	Grand Total
No	59.92%	34.82%	94.74%
Yes	4.40%	0.86%	5.26%
Grand Total	64.32%	35.68%	100.00%

1. If you choose someone at random from the population, what is the probability that they

- have no heart disease
- do a moderate amount of exercise
- do a moderate amount of exercise and have no heart disease
- do a moderate amount of exercise or have no heart disease

2. Are *Heart Disease* and *Exercise levels* mutually exclusive events?

Figure 14: ETX5900 source PDF, Week 2, page 14

Task 1: Heart Disease and Exercise Level

3. Construct a new Pivot Table for Heart Disease and Exercise Level. Change the display settings to % Row Total. Use this table and answer the following questions:
a. Write a probability statement that describes the value of 83.65% from your table?
b. Interpret this value

4. Are exercise levels and heart disease independent? Provide evidence to support your answer.

5. *A lower proportion of heart disease patients do moderate exercise compared to those who do not have heart disease*. Construct a pivot table and provide probability statements to support this.

6. *The probability of heart disease reduces with moderate exercise compared to minimum exercise*. Construct a pivot table and provide probability statements to support this.

Figure 15: ETX5900 source PDF, Week 2, page 15

Solutions for Task 1: Question 1

Count of Heart Disease	Exercise Level		
Heart Disease	Minimal	Moderate	Grand Total
No	59.92%	34.82%	94.74%
Yes	4.40%	0.86%	5.26%
Grand Total	64.32%	35.68%	100.00%

1. If you choose someone at random from the population, what is the probability that they

- have no heart disease

P(No Heart Disease) = 94.74%

- do a moderate amount of exercise

P(Moderate) = 35.68%

Figure 16: ETX5900 source PDF, Week 2, page 16

Solutions for Task 1: Question 1

Count of Heart Disease	Exercise Level		
Heart Disease	Minimal	Moderate	Grand Total
No	59.92%	34.82%	94.74%
Yes	4.40%	0.86%	5.26%
Grand Total	64.32%	35.68%	100.00%

1. If you choose someone at random from the population, what is the probability that they

- do a moderate amount of exercise and have no heart disease

P(Moderate and No Heart Disease) = 34.82%

- do a moderate amount of exercise or have no heart disease

P(Moderate or No Heart Disease)

= P(Moderate) + P(No Heart Disease) – P(Moderate and No Heart Disease)

= 35.68% + 94.74% - 34.82% = 95.6%

Figure 17: ETX5900 source PDF, Week 2, page 17

Solutions for Task 1: Question 2

Count of Heart Disease	Exercise Level		
Heart Disease	Minimal	Moderate	Grand Total
No	59.92%	34.82%	94.74%
Yes	4.40%	0.86%	5.26%
Grand Total	64.32%	35.68%	100.00%

2. Are *Heart Disease* and *Exercise levels* mutually exclusive events?

Definition of mutually exclusive events: Events that cannot occur together

If A and B are mutually exclusive, P(A and B) = 0

P(Heart Disease and Minimal) = 4.4%

P(Heart Disease and Moderate) = 0.86%

Therefore Heart Disease and Exercise are not mutually exclusive.

Figure 18: ETX5900 source PDF, Week 2, page 18

Solutions for Task 1: Question 3

Count of Heart Disease	Exercise		
Heart Disease	Minimal	Moderate	Grand Total
No	63.25%	36.75%	100.00%
Yes	83.65%	16.35%	100.00%
Grand Total	64.32%	35.68%	100.00%

Q3a. Write a probability statement to describe the highlighted value.

$$\Pr(Minimal|Heart\ Disease) = 83.65\%$$

Q3b. Interpret this value.

If you only consider heart disease patients, 83.65% of them do minimal exercise

OR

Approximately 83.65% of patients with heart disease, do minimal exercise.

Figure 19: ETX5900 source PDF, Week 2, page 19

Solutions for task 1: Question 4

Are exercise levels and heart disease independent? Provide evidence to support your answer.

P(Moderate | No Heart Disease) = 36.75%
P(Moderate | Heart Disease) = 16.35%
P(Moderate) = 35.68%
Is P(Moderate | No Heart Disease) = P(Moderate)??
Conditional probability ≠ marginal probability

Count of Heart Disease	Exercise		
Heart Disease	Minimal	Moderate	Grand Total
No	63.25%	36.75%	100.00%
Yes	83.65%	16.35%	100.00%
Grand Total	64.32%	35.68%	100.00%

Count of Heart Disease	Exercise		
Heart Disease	Minimal	Moderate	Grand Total
No	59.92%	34.82%	94.74%
Yes	4.40%	0.86%	5.26%
Grand Total	64.32%	35.68%	100.00%

Are Exercise level and heart disease independent?
Exercise level and heart disease are NOT independent
There is a relationship between exercise and heart health!

Figure 20: ETX5900 source PDF, Week 2, page 20

Solutions for Task 1: Question 5

A lower proportion of heart disease patients do moderate exercise compared to those who do not have heart disease. Construct a pivot table and provide probability statements to this statement.

P(Moderate | **No Heart Disease**) = 36.75%

P(Moderate | **Heart Disease**) = 16.35%

Count of Heart Disease	Exercise		
Heart Disease	Minimal	Moderate	Grand Total
No	63.25%	36.75%	100.00%
Yes	83.65%	16.35%	100.00%
Grand Total	64.32%	35.68%	100.00%

What does this imply?

A smaller proportion of heart disease patients do moderate exercise compared to those who do not have heart disease.

Application of what you have learned:

People with healthy hearts tend to do more exercise!

Figure 21: ETX5900 source PDF, Week 2, page 21

Solutions for Task 1: Question 6

The probability of heart disease reduces with moderate exercise compared to minimum exercise. Construct a pivot table and provide probability statements to this statement.

The condition here is exercise.
P(Heart Disease|Minimum) = 6.84%
P(Heart Disease|Moderate) = 2.41%

Count of Heart Disease	Exercise		
Heart Disease	Minimal	Moderate	Grand Total
No	93.16%	97.59%	94.74%
Yes	6.84%	2.41%	5.26%
Grand Total	100.00%	100.00%	100.00%

Application of what you have learned:
The probability of heart disease reduces with moderate exercise!

Figure 22: ETX5900 source PDF, Week 2, page 22

Task 2: Marketing (to be completed in your own time)

A company is interested in analyzing the efficacy of launching a new brand of frozen food products. They wish to study the various demographics of customers who have and have not tried their products using the ***Marketing*** worksheet in the dataset **Week 3 Workshop Data.xlsx.**

This data contains 6 variables:

- Person ID = the unique identifier of each person/respondent in the data.
- Raw Income = the income of the respondent.
- Gender = the gender (male or female) of the respondent.
- Live Alone = whether the respondent lives alone.
- Have Tried = whether the respondent has tried the new product or not.
- Income Category = the income range of the respondent.

Figure 23: ETX5900 source PDF, Week 2, page 23

Task 2: Marketing (to be completed in your own time)

Instructions:

Construct contingency tables to analyse the relationship between "Have Tried" and 3 other variables: Live Alone, Income & Gender. Use these tables to answer the following questions. Include probability statements and contextual interpretations in your answers.

a. What is the proportion of customers who live alone?
b. What is the proportion of customers who have tried the frozen food product and live alone?
c. What is the probability that a randomly selected customer earns more than $50,000?
d. What is the probability that a randomly selected customer earns more than $50,000 and has tried the frozen food product?
e. What proportion of customers who live alone have tried the frozen food product?
f. What proportion of customers who have tried the frozen food product, live alone?
g. Are there more males or females who have tried the frozen food product?

Figure 24: ETX5900 source PDF, Week 2, page 24

Solutions for Task 2a (to be completed in your own time)

What is the proportion of customers who live alone?

P(Live Alone = Yes) = 16.94%

16.94% of the customers live alone.

Count of Have Tried	Live Alone		
Have Tried	No	Yes	Grand Total
No	37.85%	4.32%	42.17%
Yes	45.21%	12.62%	57.83%
Grand Total	**83.06%**	**16.94%**	**100.00%**

Figure 25: ETX5900 source PDF, Week 2, page 25

Solutions for Task 2b (to be completed in your own time)

What is the proportion of customers who have tried the frozen food product and live alone?

P(Have Tried = Yes AND Live Alone = Yes) = 12.62%

12.62% of the customers have tried the frozen food and live alone.

Count of Have Tried	Live Alone		
Have Tried	**No**	**Yes**	**Grand Total**
No	37.85%	4.32%	42.17%
Yes	45.21%	12.62%	57.83%
Grand Total	**83.06%**	**16.94%**	**100.00%**

Figure 26: ETX5900 source PDF, Week 2, page 26

Solutions for Task 2c (to be completed in your own time)

What is the probability that a randomly selected customer earns more than $50,000?

P(Income > 50,000) = 21.14% + 7.24% + 5.84% = 34.23%
34.23% of the customers earn more than $50,000.

Count of Have Tried	Have Tried		
Income	**No**	**Yes**	**Grand Total**
<25,000	12.62%	12.73%	25.35%
25,000-50,000	17.29%	23.13%	40.42%
50,000-75,000	8.06%	13.08%	21.14%
75,000-100,000	2.57%	4.67%	7.24%
over 100,000	1.64%	4.21%	5.84%
Grand Total	**42.17%**	**57.83%**	**100.00%**

Figure 27: ETX5900 source PDF, Week 2, page 27

Solutions for Task 2d (to be completed in your own time)

What is the probability that a randomly selected customer earns more than $50,000 and has tried the frozen food product?

P(Income > 50,000 AND Have Tried = Yes) = 13.08% + 4.67% + 4.21% = 21.96%

21.96% of the customers earn more than $50,000 and has tried the frozen food product.

Count of Have Tried	Have Tried		
Income	**No**	**Yes**	**Grand Total**
<25,000	12.62%	12.73%	25.35%
25,000-50,000	17.29%	23.13%	40.42%
50,000-75,000	8.06%	13.08%	21.14%
75,000-100,000	2.57%	4.67%	7.24%
over 100,000	1.64%	4.21%	5.84%
Grand Total	**42.17%**	**57.83%**	**100.00%**

Figure 28: ETX5900 source PDF, Week 2, page 28

Solutions for Task 2e (to be completed in your own time)

What proportion of customers who live alone have tried the frozen food product?

P(Have Tried = Yes | Live Alone = Yes) = 74.48%
74.48% of customers who live alone, have tried the frozen food.

Count of Have Tried	Live Alone		
Have Tried	**No**	**Yes**	**Grand Total**
No	45.57%	25.52%	42.17%
Yes	54.43%	74.48%	57.83%
Grand Total	**100.00%**	**100.00%**	**100.00%**

Figure 29: ETX5900 source PDF, Week 2, page 29

Solutions for Task 2f (to be completed in your own time)

What proportion of customers who have tried the frozen food product, live alone?

P(Live Alone = Yes | Have Tried = Yes) = 21.82%

21.82% of customers who have tried the frozen food product, live alone.

Count of Have Tried	Live Alone		
Have Tried	**No**	**Yes**	**Grand Total**
No	89.75%	10.25%	100.00%
Yes	78.18%	21.82%	100.00%
Grand Total	**83.06%**	**16.94%**	**100.00%**

Figure 30: ETX5900 source PDF, Week 2, page 30

Solutions for Task 2g (to be completed in your own time)

Are there more males or females who have tried the frozen food product?

P(Male | Have Tried = Yes) > P(Female | Have Tried= Yes) 56.36% > 43.64%
Therefore, there are more males who have tried the frozen food product than females.

Count of Have Tried	Have Tried		
Gender	**No**	**Yes**	**Grand Total**
Female	50.42%	43.64%	46.50%
Male	49.58%	56.36%	53.50%
Grand Total	**100.00%**	**100.00%**	**100.00%**

Figure 31: ETX5900 source PDF, Week 2, page 31

Motherboard

Digital World manufactures motherboards and has factories in China and Australia. Information about their production is provided in the ***Motherboard*** worksheet in the dataset ***Week 2 Workshop Data.xlsx.***

The company wishes to investigate the quality of the motherboards being manufactured from each of the factories.

Use the following notation in all your answers:

D: a motherboard is defective *G: a motherboard is good (not defective) C: motherboard manufactured in China* *A: motherboard manufactured in Australia*

Figure 32: ETX5900 source PDF, Week 2, page 32

Challenge

You are encouraged to work in groups.





Figure 33: ETX5900 source PDF, Week 2, page 33

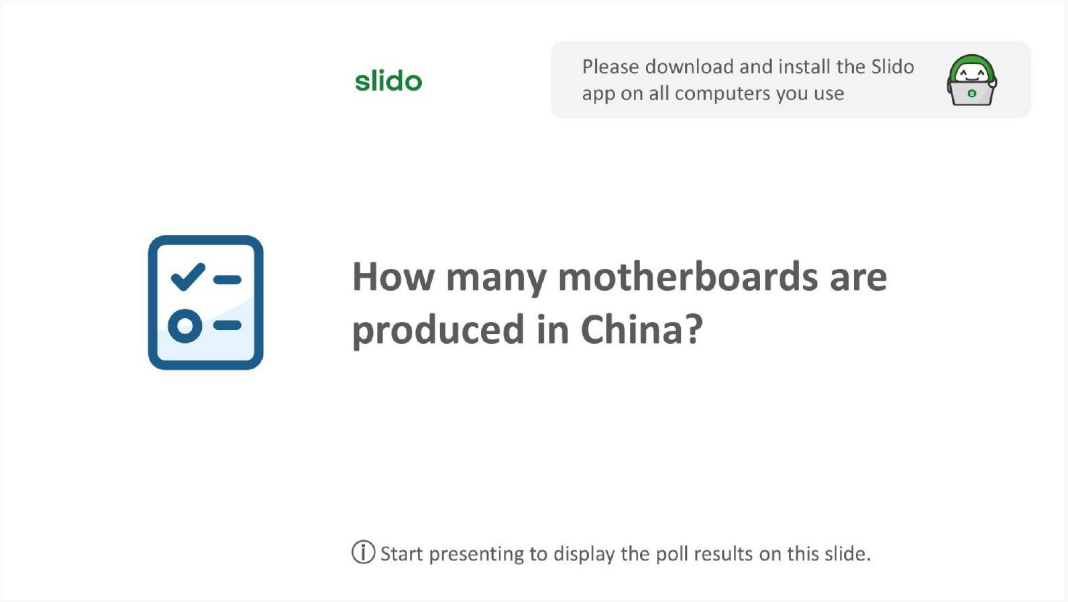


Figure 34: ETX5900 source PDF, Week 2, page 34

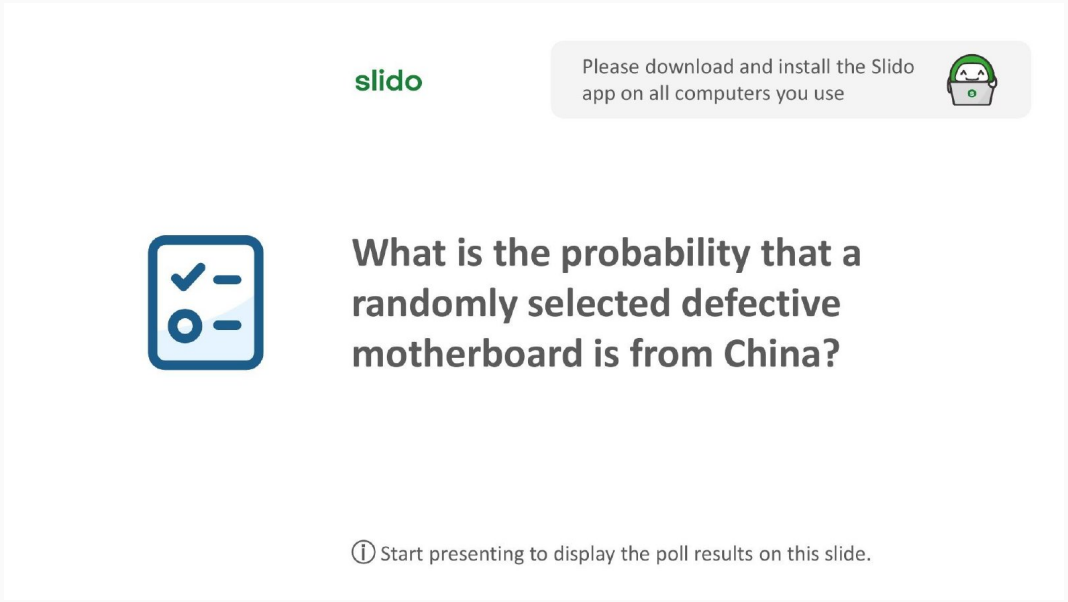


Figure 35: ETX5900 source PDF, Week 2, page 35

Solutions for poll questions

How many motherboards are produced in China? 65

What is the probability that a randomly selected defective motherboard is from China?

P(C|D) = 33.33%

33.33% of defective motherboards are from China.

Count of Quality Of Products	Quality Of Products		
Production of Motherboards	**Defective**	**Good**	**Grand Total**
Australia	66.67%	32.98%	35.00%
China	33.33%	67.02%	65.00%
Grand Total	**100.00%**	**100.00%**	**100.00%**

Figure 36: ETX5900 source PDF, Week 2, page 36

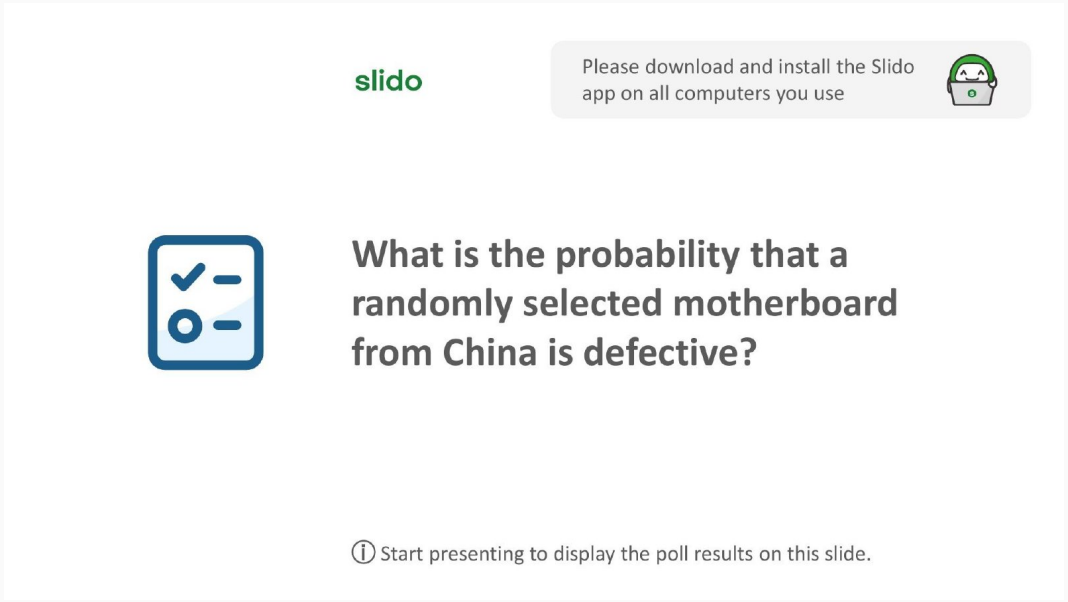


Figure 37: ETX5900 source PDF, Week 2, page 37

Solutions for poll questions

What is the probability that a randomly selected motherboard from China is defective?

$P(D|C) = 3.08\%$

3.08% of motherboards from China are defective.

Count of Quality Of Products	Quality Of Products		
Production of Motherboards	Defective	Good	Grand Total
Australia	11.43%	88.57%	100.00%
China	3.08%	96.92%	100.00%
Grand Total	**6.00%**	**94.00%**	**100.00%**

Figure 38: ETX5900 source PDF, Week 2, page 38

Part 2 of the workshop: Normal Distribution

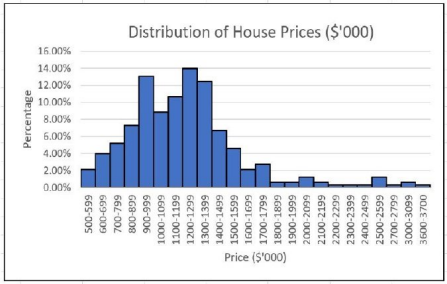
- Probability: as the area under a Probability density function (*pdf*)
- The normal distribution: shape, location and distribution.
- How to specify a particular normal distribution
- The effect of changing the mean and standard deviation
- Finding areas under the standard normal curve: Using Excel

40

Figure 39: ETX5900 source PDF, Week 2, page 39

Probability Distribution

- The histogram is a picture of the probability density in crude form
- Probability **density** describes how "**dense**" the distribution is over a data range
- Allows us to **calculate the probability** related to the variable of interest
- In statistics, we use a **smooth mathematical function** to model the **probability density function (pdf)**
- **The area under the curve represents the probability**



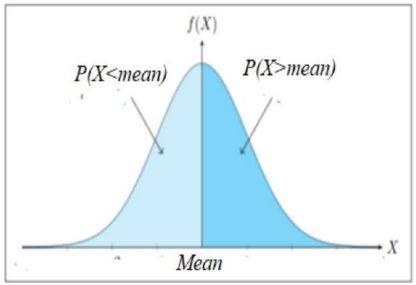
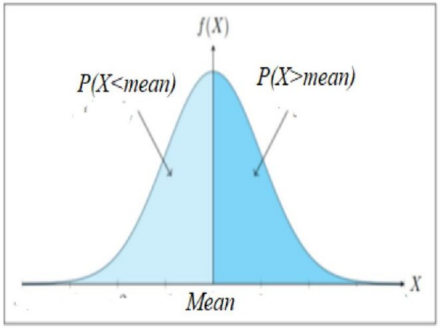


Figure 40: ETX5900 source PDF, Week 2, page 40

Probability Distribution: Normal Distribution

- The most common distribution in statistics is the **normal distribution**
- It is a **symmetric (bell-shaped)** distribution
- The normal distribution has two features: Mean and Stdev



Normal Distribution

- **Notation:**

  $\boldsymbol{X \sim N(Mean, Stdev)}$

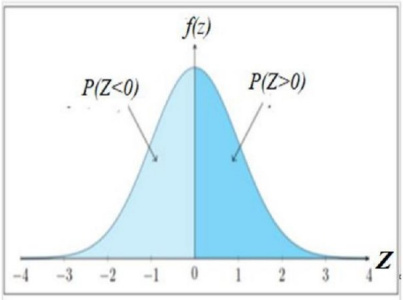
  $\boldsymbol{X \sim N(\mu, \sigma)}$

- Skewness = 0
- Mean = Median = Mode

Figure 41: ETX5900 source PDF, Week 2, page 41

STANDARD Normal Distribution

- A special case: **STANDARD normal distribution**
- Mean = 0 and Stdev = 1
- Used in statistics to assess statistical uncertainty

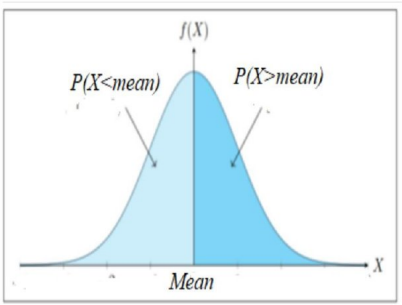


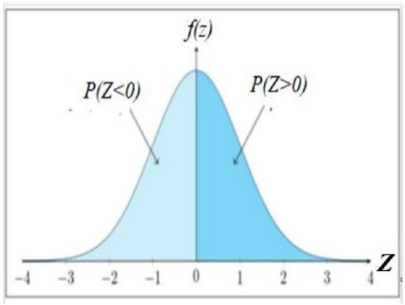
Standard Normal Distribution

- Notation: $Z \sim N(0,1)$
- Skewness = 0
- Mean = Median = Mode

Figure 42: ETX5900 source PDF, Week 2, page 42

Normal Distribution & Standard normal distribution





$X \sim N(Mean, Stdev)$

$Z \sim N(0,1)$

Figure 43: ETX5900 source PDF, Week 2, page 43

Standard normal distribution: Excel

Question 1: $P(Z<0)$

Question 2: $P(Z>0)$

Question 3: $P(Z<-1)$

Question 4: $P(-1<Z<1.5)$

Question 5: $P(Z>2)$

Question 6: Suppose Z has a standard normal distribution. For some value of $z*$, the probability $P(Z<z*)=0.2843$. Find $z*$.

Question 7: Suppose Z has a standard normal distribution. For some value of $z*$, the probability $P(Z>z*)=0.2843$. Find $z*$.

Figure 44: ETX5900 source PDF, Week 2, page 44

Standard normal distribution: Excel

Question 1	P(Z<0)	=NORM.S.DIST(0,TRUE)	**0.5**
Question 2	P(Z>0)	=1-NORM.S.DIST(0,TRUE)	**0.5**
Question 3	P(Z<-1)	=NORM.S.DIST(-1,TRUE)	**0.158655254**
Question 4	P(-1<Z<1.5)	=NORM.S.DIST(1.5,TRUE)-NORM.S.DIST(-1,TRUE)	**0.0.774538**
Question 5	P(Z>2)	=1-NORM.S.DIST(2,TRUE)	**0.02275**

Figure 45: ETX5900 source PDF, Week 2, page 45

Standard normal distribution: Excel

Question 6
Suppose Z has a standard normal distribution. For some value of $z*$, the probability $P(Z<z*)=0.2843$. Find $z*$.

To find z*
=norm.s.inv(0.2843) = -0.57011



Question 7
Suppose Z has a standard normal distribution. For some value of $z*$, the probability $P(Z>z*)=0.2843$. Find $z*$.
=Norm.s.inv(1-0.2843) = 0.57011

Figure 46: ETX5900 source PDF, Week 2, page 46

Checklist: General Probability & Normal Distribution

- **Use contingency table to calculate marginal, joint and conditional probabilities**
  - % of Grand Total
  - % of Column Total
  - % of Row Total
- **Interpreting the relationship between two categorical variables**
  - Rule of Independence
- **Probability: as the area under a Probability density function (pdf)**
- **The normal distribution: shape, location and distribution.**
- **How to specify a particular normal distribution**
- **The effect of changing the mean and standard deviation**
- **Finding areas under the standard normal curve: Using Excel**

Figure 47: ETX5900 source PDF, Week 2, page 47